



Monomial maps and toric ideals

$$A = (1 \ 2) \in \mathbb{Z}^{1 \times 2}$$

$$\phi_A: \mathbb{C} / \{0\} \rightarrow \mathbb{C}^2$$

$$t \mapsto (t^1, t^2)$$

Affine toric variety:

$$A \in \mathbb{Z}^{d \times n}$$

$$\phi_A: (\mathbb{C}^*)^d \rightarrow \mathbb{C}^n$$

$$(t_1, \dots, t_d) \mapsto (t_1^{a_1}, \dots, t_d^{a_n})$$

$$Y_A = \overline{\text{im}(\phi_A)}$$

Torus :

A d -dim torus is an algebraic variety
isomorphic to $(\mathbb{C}^*)^d$

Recap:

Affine semigroups:

$$S = NA = \left\{ \sum_{a \in A} c_a a : c_a \in \mathbb{N} \right\} \\ \subseteq \mathbb{Z}^d$$

Semigroup algebra:

$$\mathbb{C}[S] = \left\{ \sum_{u \in S} c_u t^u : c_u \in \mathbb{C}, \right. \\ \left. \text{finitely many } c_u \text{ non-zero} \right\} \\ \subseteq \mathbb{C}[t_1^{\pm 1}, \dots, t_d^{\pm 1}]$$

Rational polyhedral cones:

$$\sigma = \text{cone}(S) = \left\{ \sum_{u \in S} \lambda_u u : \lambda_u \in \mathbb{R}_{\geq 0} \right\}$$

Convex lattice polytopes

A convex polytope $\mathcal{P}$ in $M_{\mathbb{R}}$ is the convex hull

$$\mathcal{P} = \text{conv}(A) = \left\{ \sum_{i=1}^n c_i a_i \in M_{\mathbb{R}} : c_i \in \mathbb{R}, \sum c_i = 1, c_i \geq 0 \right\} \subset M_{\mathbb{R}}$$

Define hyperplanes: $u \in M_{\mathbb{R}} \setminus \{0\}, b \in \mathbb{R}$

$$H_{u,b} = \{ m \in M_{\mathbb{R}} : \langle u, m \rangle + b = 0 \}$$

$$H_{u,b}^+ = \{ m \in M_{\mathbb{R}} : \langle u, m \rangle + b \geq 0 \}$$

A face of a polytope $\mathcal{P}$ is the set $H_{u,b} \cap \mathcal{P}$

if $\mathcal{P} \subset H_{u,b}^+$ and $b = -\min_{m \in \mathcal{P}} \langle u, m \rangle$.

Denote the face as $Q \leq \mathcal{P}$. $H_{u,b}$ is a supporting hyperplane.

The codimension is $\text{codim}(Q) = \dim \mathcal{P} - \dim Q$

Examples :

V-representation :

$$\mathcal{P} = \text{Conv} \{ v \in M_{\mathbb{R}} : v \text{ is vertex of } \mathcal{P} \}$$

H-representation :

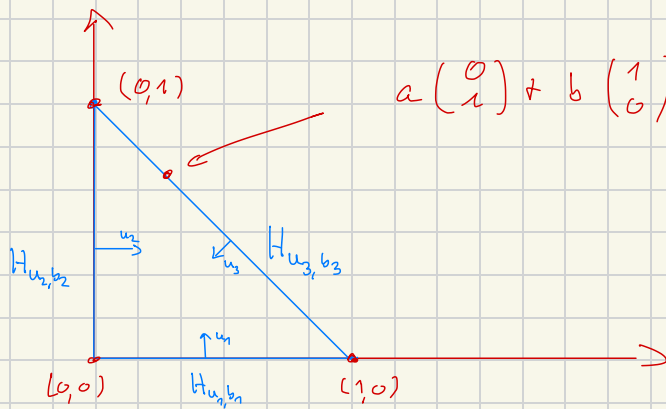
$$\mathcal{P} = \bigcap_{i=1}^R H_{u_i, b_i}^+ \quad \text{with } H_{u_i, b_i} \text{ supp. hyperplanes}$$

Facet normal : Let $\mathcal{P}$ be full-dimensional
then for a facet $Q \subset \mathcal{P}$ we
can choose u_Q as generator
of

$$\{ u \in N_{\mathbb{R}} : \langle u, m \rangle = 0 \quad \forall m \in Q \}$$

Then $\mathcal{P} \subset H_{u_Q, b_Q}^+ \Rightarrow u_Q$ is per-p. to Q ,
pointing inward.

Example: $M_{\mathbb{R}^2} = \mathbb{R}^2$



$$a \begin{pmatrix} 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad a+b=1, \quad a, b \geq 0$$

$$H_{u_2, b_2} = \left\{ m \in \mathbb{R}^2 : \langle u_2, m \rangle + b_2 = 0, \quad u_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

$$H_{u_3, b_3} = \left\{ m \in \mathbb{R}^2 : \langle u_3, m \rangle + b_3 = 0, \quad u_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

$$H_{u_1, b_1} = \left\{ m \in \mathbb{R}^2 : \langle u_1, m \rangle + b_1 = 0, \quad u_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

A dilation of a polytope $\mathcal{P} \subset M_{\mathbb{R}^2}$ by $\lambda \in \mathbb{R}_{\geq 0}$ is the polytope $\lambda \cdot \mathcal{P} = \{ \lambda p : p \in \mathcal{P} \}$

For a lattice polytope we have:

Theorem (Ehrhart)

$P \subset \mathbb{R}^n$ convex lattice polytope, $\dim P = d$.

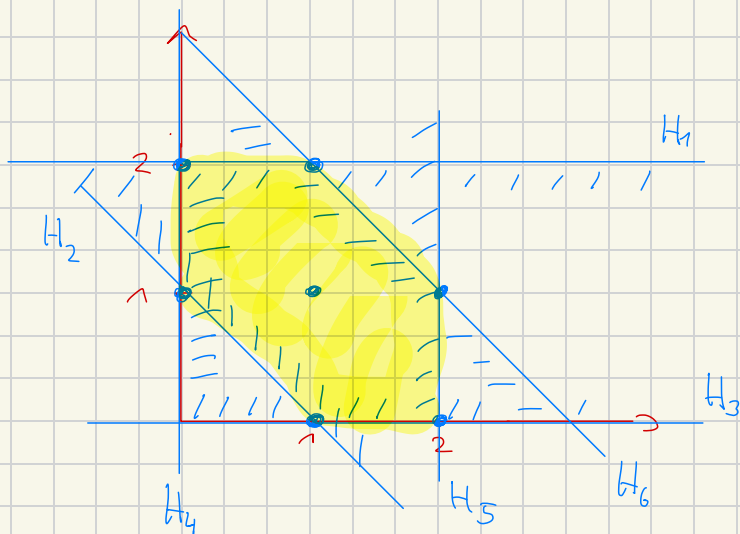
Define: $E_P: \mathbb{N} \rightarrow \mathbb{N}$ (Ehrhart polynomial)

$$k \mapsto |k \cdot P \cap \mathbb{Z}^n|$$

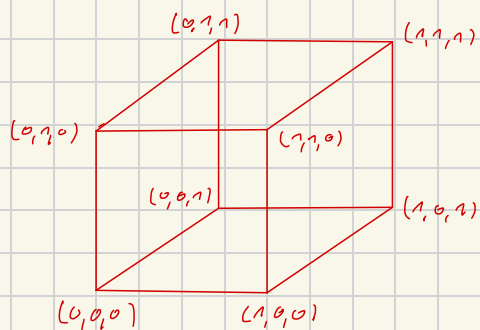
The leading term of the polynomial $E_P(k)$ is $\text{Vol}(P)k^d$, with

$$\text{Vol}(P) = \int_P dx_1 \dots dx_d$$

Exercise 3.1.11: $A' = \begin{pmatrix} 1 & 0 & 0 & 1 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 1 & 0 & 1 \end{pmatrix}$



Exercise 3.1.12 :



Affine charts of a projective toric variety

Let $U_i = \mathbb{P}^{n-1} \setminus V(x_i)$ be the affine chart for $\mathbb{P}^{n-1}$.

We have an isomorphism $U_i \cong \mathbb{C}^{n-1}$ given by

$$\mathcal{Z}_i : (x_1 : \dots : x_n) \mapsto \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right)$$

$$\text{For } \phi_A : (\mathbb{C}^*)^d \rightarrow \mathbb{P}^{n-1}, \quad A \in \mathbb{Z}^{d \times n} \\ t = t_1 \dots t_d \mapsto (t^{a_1} : \dots : t^{a_n})$$

We have the composition with $\mathcal{Z}_i$:

$$\phi_{A-a_i} = \mathcal{Z}_i \circ \phi_A : (\mathbb{C}^*)^d \rightarrow (\mathbb{C}^*)^{n-1}$$

$$\text{given by: } t \mapsto (t^{a_1-a_i}, \dots, t^{a_{i-1}-a_i}, t^{a_{i+1}-a_i}, \dots, t^{a_n})$$

$$\text{We get an affine covering: } X_A = \bigcup_{i=1}^n (X_A \cap U_i)$$

Prop. : Each affine chart $X_A \cap U_i$ is isomorphic to Y_{A-a_i} with

$$A - a_i = (a_n - a_i, \dots, a_{i-1} - a_i, a_{i+1} - a_i, \dots, a_n)$$

$\Rightarrow$ We have $\varphi_i(X_A \cap U_i) \subset \text{im } \Phi_{A-a_i} \subset \mathbb{C}^{n-1}$

and in fact

$$\overline{\text{im } \Phi_{A-a_i}} = \varphi_i(X_A \cap U_i) = Y_{A-a_i}$$

Exercise 3.2.2.

Projective closure of $\mathbb{C}^n$

$$g: \mathbb{C}^n \hookrightarrow \mathbb{P}^n$$

$$(z_1, \dots, z_n) \mapsto (1: z_1: \dots: z_n)$$

For Y_A we have $Y_A = \overline{\text{im } \Phi_A} = \overline{\{ \Phi_A(t) : t \in (\mathbb{C}^*)^d \}} \subset \mathbb{C}^n$

Define $\overline{Y}_A$ the projective closure of Y_A .

$$\text{So: } \overline{Y}_A = \overline{\{ (1: t^{a_1}: \dots: t^{a_n}) : t \in (\mathbb{C}^*)^d \}} \subset \mathbb{P}^n$$

But this is exactly the definition of $X_{\bar{A}}$ for $\bar{A} = \{0\} \cup A \in \mathbb{Z}^{d \times (n+1)}$

Injective projective varieties:

Define $L = 0 \cup (A - a_1) \in \mathbb{Z}^{d \times n}$ and $P \perp Q = S$ ↙ works for any column a_i .

the SNF. Define $r = \text{rank}(L)$ and

$\bar{L} \in \mathbb{Z}^{r \times n}$ the matrix of rows of Q^{-1} .

Then $X_A = X_L = X_{\bar{L}}$ and

$$\Phi_{\bar{L}}: (\mathbb{C}^*)^d \rightarrow \text{im } \Phi_{\bar{A}}$$

is injective (one-to-one)

Proof: $X_A = X_{\bar{A}} = X_L$ (Prop. 13.14) since $A \rightarrow L \rightarrow \bar{L}$ are affine transformations.

We have that $\mathcal{Z}_1 = \bar{\Phi}_{\bar{A}} = \Phi_{\bar{A} - \bar{a}_1}$. But $\Phi_{\bar{A} - \bar{a}_1}$ is injective and $\mathcal{Z}_2$ is as well. So $\Phi_{\bar{L}}$ must be injective

Minimal affine covering:

$X_A = \bigcup_i (X_A \cap U_i)$ is redundant. The minimal covering can be found by defining

$$V = \{i : a_i \in A \text{ is a vertex of } \text{Conv}(A)\}$$

Then

$$X_A = \bigcup_{i \in V} (X_A \cap U_i)$$

Proof:

